

SUMIN PARK

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 [Google Scholar](#) |  [LinkedIn](#) |  [Github](#)

EDUCATION

- **Korea Advanced Institute of Science and Technology (KAIST)** Sep 2024 – Present
M.S. in Computer Science Daejeon, South Korea
 - GPA: 4.17 / 4.30
- **Ewha Womans University** Mar 2016 – Feb 2021
B.S. in Brain and Cognitive Sciences & B.S. in Life Sciences Seoul, South Korea
 - GPA: 4.07 / 4.30 (Summa Cum Laude)
 - Thesis: *Differential Mechanism of Positive and Negative Valence Representation: A Focus on Amygdalar Circuitry*

EXPERIENCE

- **CAMAIL, Max Planck Institute for Intelligent Systems / ELLIS Institute Tübingen** July 2026 – Dec 2026
Researcher Intern Tübingen, Germany
 - Advisor: T. Konstantin Rusch
 - Research Topic: Blockwise parallelizable linear recurrences for efficient sequence modeling.
- **Big Data Analysis & Learning Laboratory (BigDyL), KAIST** Sep 2024 – Present
Master's Student Researcher Daejeon, South Korea
 - Advisor: Noseong Park
 - Research Topic: Interpretable LLM architectures, State-Space-Models (SSMs), Mixture-of-Expert (MoE), Representation learning, Brain-inspired AI.
- **Big Data Analysis & Learning Laboratory (BigDyL), Yonsei University** Jul 2023 – Aug 2024
Research Intern Seoul, South Korea
 - Advisor: Noseong Park
 - Research Topic: Graph representation learning, Graph neural networks (GNN)
- **DNI Consulting** Feb 2022 – Apr 2024
Data Scientist Seoul, South Korea
 - Research Topic: Data analysis and modeling for B2B marketing and customer segmentation
- **Cell Physiology Laboratory, Seoul National University** Aug 2020 – Jan 2021
Undergraduate Research Intern Seoul, South Korea
 - Advisor: Seokho Lee
 - Research Topic: Electrophysiology, PFC-involved systems consolidation
- **Neurobiology Laboratory, Seoul National University** Dec 2019 – Feb 2020
Undergraduate Research Intern Seoul, South Korea
 - Advisor: Bongkiun Kaang
 - Research Topic: Molecular analysis on dopaminergic reward circuitry

PUBLICATIONS

INTERNATIONAL CONFERENCES

- [C7] **Looped Reasoning Finishes Earlier Than You Think** | *Preprint.*
 - [Sumin Park](#), and T. Konstantin Rusch
- [C6] **Q-Delta: Beyond Key-Value Associative State Evolution** | *ICML 2026*
 - [Sumin Park](#), Seojin Kim, and Noseong Park
- [C5] **STAR: Rethinking MoE Routing as Structure-Aware Subspace Learning** | *ICML 2026*
 - [Sumin Park](#) and Noseong Park
- [C4] **How Many Experts Are Enough? Towards Optimal Semantic Specialization for Mixture-of-Experts** | *AAAI 2026*
 - [Sumin Park](#) and Noseong Park

[C3]
Are Graph Transformers Necessary? Efficient Long-Range Message Passing with Fractal Nodes in MPNNs
| AAAI 2026 (Oral)

- Jeongwhan Choi, Seungjun Park, **Sumin Park**, Sung-Bae Cho and Noseong Park

[C2]
DARS: Robust Sparse Fine-Tuning with Regularized Subspace Disalignment
| ICLR 2025 Workshop (SCOPE)

- Sumin Park** and Noseong Park

[C1]
PANDA: Expanded Width-Aware Message Passing Beyond Rewiring
| ICML 2024

- Jeongwhan Choi, **Sumin Park**, Hyowon Wi, Sung-Bae Cho, and Noseong Park

INDUSTRIAL PROJECTS

• Simulation-Based Defect Variable Identification from High-Resolution Images

Mar 2025 – Dec 2025

- Ongoing **Samsung-KAIST** industry-academia project
- Developed a multi-task, multi-class image classification framework to predict defective variables from simulation-generated images
- Designed a ViT architecture with hierarchical attention for efficient high-resolution image processing, achieving 99.41% classification accuracy

LEADERSHIP AND EXTRACURRICULARS

• Ewha Institute of Brain & Cognitive Science, Ewha Womans University

Spring 2020 – Fall 2020

President of Biomedical Division of Institute

• Debate Association of Ewha (DAE), Ewha Womans University

Spring 2019 – Fall 2019

Regular Member

- Competed in national debate tournaments (SRT, KNC); SRT 2019 Quarter Finalist

HONORS AND AWARDS

- Summa Cum Laude, Ewha Womans University2016 – 2020
- Dean’s List (7 semesters), Ewha Womans University2016 – 2020
- Academic Honors Scholarship, Ewha Womans UniversitySpring 2020
- Global Leader in Major, Ewha Womans UniversitySpring 2020
- Academic Honors Scholarship, Ewha Womans UniversitySpring 2019
- Admissions Scholarship (Full Tuition, 2 Years), Ewha Womans University2016 – 2018

SKILLS

- PROGRAMMING & FRAMEWORKS:** Python, R, C++, PyTorch, Triton
- SYSTEMS & TOOLS:** Linux, Git, Docker, AWS, SQL